BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ABBOTTABAD

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022 • Paper Code: 9221

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** 9221

Note: Attempt all questions of Section-A by filling the corresponding bubble on the MCQs RESPONSE SHEET. It is mandatory to return the attempted MCQs sheet to the Superintendent within given time.

Q1. Which one of the following has the same family as fluorine?

(A) silicon
(B) iodine
(C) antimony
(D) arsenic

Q2. Which of the following is polar?

(A) CO
(B) H_2O
(C) HCl
(D) all of these

Q3. Which of the following can flow?

(A) solid
(B) liquid
(C) gas
(D) both B&C

Q4. Water droplet in air is an example of solution .....

(A) gas in gas
(B) gas in liquid
(C) liquid in gas
(D) liquid in liquid

Q5. Elements of Group-IIA are called .....

(A) alkali metals
(B) alkaline earth metals
(C) halogens
(D) chalcogen

Q6. The f sub-shell has ..... orbital.

(A) two
(B) three
(C) five
(D) seven

Q7. Electron in its ground state does not .....

(A) spin
(B) revolve
(C) radiate energy
(D) reside in orbit

Q8. The number of neutrons in deuterium is .....

(A) zero
(B) one
(C) two
(D) three

Q9. _____ unit is the smallest repeating unit of a/an ..... compound.

(A) ionic
(B) covalent
(C) [illegible in source scan]
(D) metallic

Q10. The Chemical name of bleaching powder is .....

(A) sodium bicarbonate
(B) sodium carbonate
(C) caustic soda
(D) calcium hypochlorite

Q11. The maximum number of electrons in third orbit is .....

(A) 2
(B) 8
(C) 18
(D) 32

Q12. Sixth period contains ..... elements.

(A) 30
(B) 32
(C) 10
(D) 18

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 32 | **Paper Code:** 9221

Q1. Attempt any eight of the following short questions. Each question carries 4 marks

(i) Calculate the number of molecules in 6.50 mol of CH_4.

(ii) Differentiate between industrial and physical chemistry.

(iii) Define and draw the structures of hydrogen isotopes.

(iv) What are the trends of ionization energy in groups and periods?

(v) Define metallic bond with examples.

(vi) Differentiate between evaporation and condensation.

(vii) How can we easily distinguish saturated, unsaturated and super saturated solution from each other?

(viii) Why is NaOH strong but NH_4OH weak electrolyte?

(ix) How does ionization energies value vary in group?

(x) Define mole with two examples.

(xi) Enumerate the factors affecting electronegativity.

SECTION-C

Maximum Marks: 21

Note: Attempt any three of the following questions. Each question carries 7 marks

Q2.

(i) State the postulates of Bohr's atomic theory.

(ii) Draw the Lewis structures of CO, CCl_4, SO_2 and HCl.

Q3.

(i) Define Boyle's law and verify it experimentally.

(ii) Write any three differences between compound and mixture.

Q4.

(i) Compare colloids and suspension.

(ii) Calculate the molarity of a solution containing 7.5 mol of $CaCO_3$ in enough water to make 1.50 dm^3 of solution.

Q5.

(i) Highlight the main distinction between ionic and covalent bond?

(ii) What happens during displacement reaction in halogens?